

Final Internship Report — UPRO Soccer

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Main Workstreams:

1. **Data Labeling** (video → structured JSON labels)
 2. **AI Text Response** (metrics → scoring/rubric → coaching feedback API)
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1. Executive Summary

During my internship at UPRO Soccer, I worked on two core parts of the product pipeline that powers personalized training feedback for youth soccer players:

- **Workstream 1 — Data Labeling:** I labeled drill videos (mostly Toe Tap drills) using event-based annotations tied to pose estimation/object detection. The deliverables were exported as **JSON files** and submitted to the team's shared Google Drive.
- **Workstream 2 — AI Text Response:** I contributed to the design and implementation of a production-ready feedback generator that converts drill analytics into **6 standardized UPRO metrics**, interprets performance with a rubric, compares against **historical trends**, and returns **age-appropriate coaching feedback** via a FastAPI service.

A major outcome of the internship was moving from an early **single-script demo pipeline** to a **team-built production-style API** with strong structure: multi-drill sessions, tiered feedback, trend analysis, drill recommendations, and test/debug endpoints.

2. Project Context and Goal

UPRO Soccer is building an experience where a player completes training drills, the system analyzes performance signals (pose/ball tracking and derived metrics), and the app returns feedback that is:

- **personalized**
- **age-appropriate** (kid-friendly to coach-professional)
- **actionable** (what to improve + which drill to practice next)

The internship goal was to support this end-to-end pipeline by strengthening:

1. the **quality of the data inputs** (labels), and
 2. the **reliability and usability of the feedback outputs** (text feedback system).
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Workstream 1 — Data Labeling

3. Objective

Create high-quality event-based labels for drill videos to support analytics generation and downstream feedback.

4. Labeling Scope

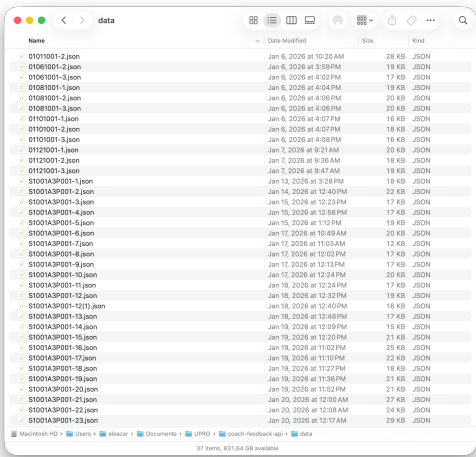
- Primary drill type: **Toe Taps** (and variations depending on video)
- Label types: **frame/event-based annotations** aligned with pose estimation/object detection workflows
- Output format: **JSON labeling exports**, consistent naming and structure for team use

5. Deliverables and Evidence

All deliverables were uploaded to:

Google Drive → UPRO - AI PACE / Humberto

- Based on the folder evidence, the deliverables include:
- **At least ~36 JSON files** visible in the folder listing (total may be higher)
- **Date range: Jan 6, 2026 → Jan 20, 2026**
- **Naming patterns:**
 - early batches like 01061001-1/2/3.json
 - main batch like S1001A3P001-1.json up to S1001A3P001-23.json (including one duplicate version S1001A3P001-12(1).json)
- **Typical file size:** ~12 KB – 29 KB each (consistent with structured labeling exports)



6. Impact

This labeled dataset improves the pipeline by:

- enabling consistent metric extraction (reducing missing/ambiguous events),
- supporting reproducibility across sessions and players,
- giving the team a reliable foundation for drill analytics and feedback generation.

Workstream 2 — AI Text Response System (Primary Technical Contribution)

7. Objective

Build a system that transforms drill analytics into meaningful coaching feedback by:

1. extracting and standardizing scores,
2. interpreting them consistently using a rubric,
3. comparing performance against recent history, and
4. returning **fast + readable feedback** suited to the player’s age and skill level.

8. Team Solution Overview

The team's repository (**upro-feedback-generator**) is a production-style service built around **FastAPI**, designed to be used by the app or other services.

1. **What the service does (simple explanation)**
2. **Receives session JSON:** player info + multiple drills with analytics
3. **Extracts 6 UPRO metric scores per drill**
4. **Averages scores across drills** for the session
5. **Compares against the last 3 sessions** (trend analysis)
6. **Finds strengths & weaknesses** using a rubric
7. Returns feedback in two modes:
 - **Summary** (instant templates, no API calls)
 - **Detailed** (LLM-generated feedback + trends + drill recommendations)

9. Key Features Implemented in the Team Repo

A) Multi-drill session support

Instead of assuming one drill, the system accepts **multiple drills per session** and outputs:

- **per-drill scores**
- **averaged session scores**
- session metadata (date, count of drills)

This matches real user behavior (players complete multiple drills).

B) Tiered feedback system (3 tiers)

The system is intentionally designed to be reliable and fast:

- **Tier 1 — Scores:** Extracted numeric scores (per drill + average)
- **Tier 2 — Summary feedback:** Hardcoded templates (instant, ~<50ms, no LLM)
 - Uses age groups (≤ 8 , 9–12, ≥ 13)
 - Random template selection for variety
 - Logic: best metric $\rightarrow$ compliment; worst metric $\rightarrow$ improvement
 - If all scores are strong (>80), return only compliment
- **Tier 3 — Detailed feedback:** OpenAI-generated coaching feedback
 - Uses history comparison and weaknesses
 - Produces a structured JSON response with:
 - trend observation
 - drill-by-drill insight
 - key improvement area
 - one actionable drill

This tiering is important because it provides a “safe default” without external dependencies, and a richer option when needed.

C) Age-appropriate coaching

The system adjusts language and coaching style based on age group:

- **≤ 8 :** simple, fun, encouraging (no emojis for mobile display)
- **9–12:** clear, balanced feedback, basic technical terms
- **≥ 13 :** professional coaching language, benchmarks, match scenarios

D) Trend analysis (history-aware feedback)

A major improvement is that feedback isn't only about "today"; it also considers progress.

- Uses the **last 3 sessions** (currently hardcoded for testing, planned to be DB-backed later)
- Calculates:
 - **session trend** (improving/declining/consistent)
 - **metric trends** (which metrics improved or declined)
 - **drill trends** (each drill vs historical performance)

E) Actionable drill recommendations

A drill database maps weaknesses to practice drills, filtered by skill level difficulty:

- Beginner/Recreational → easier drills
- Developmental/Intermediate → moderate difficulty
- Pro/Advanced → higher difficulty

This makes feedback actionable and supports "next best drill" inside the app.

10. How the Service Works

The team's **FastAPI endpoint** POST /api/v1/feedback orchestrates the pipeline:

1. **Validate request** using Pydantic schemas
2. **Extract player info** (age, skill level)
3. **Extract per-drill scores** from drill analytics
4. **Average scores** across drills
5. **Build history comparison** (last 3 sessions)
6. **Interpret rubric + strengths/weaknesses**
7. If feedback_type="summary" → templates (Tier 2)
8. If feedback_type="detailed" → OpenAI prompt builder + structured JSON output (Tier 3)

The repo also includes:

- /health for deployment readiness
 - /docs + /redoc for API documentation
 - debug endpoints to test extraction + switch trend profiles
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My Approach vs Team Approach

11. My earlier repo (discontinued)

My repo (**coach-feedback-api**) was an earlier **demo pipeline** focused on proving the concept end-to-end:

- single CLI script style
- took labeling exports + session outputs
- computed low-level metrics (e.g., touches/min, rhythm, left/right balance)
- used thresholds + rubric to generate feedback
- optional OpenAI call and optional LangSmith logging

It was useful to validate ideas quickly, but it was not optimized for production integration.

12. Team repo

The team's approach is stronger because it is designed like a real product service:

Architecture

- Production-style **FastAPI API** (/api/v1/feedback)
- Standard schemas (Pydantic), tests, docker structure
- Multi-drill sessions and consistent response format

Feedback quality

- Standardized 6 UPRO metrics (stable contract for the app)
- Tiered feedback for reliability (summary always works)
- Age-specific style and structured output

Personalization

- Trend analysis across last 3 sessions
- Drill-by-drill insights
- Actionable drill recommendations from a drill library

My earlier pipeline was a fast prototype; the team repo is an actual deployable service that supports the product roadmap.

Testing, Reliability, and Deployability

13. Testing and Debugging Support

The team repo includes:

- example request payloads for multiple realistic scenarios (improving, declining, fatigue, teen/young player)
- debug endpoints for:
 - checking extraction output
 - switching history profiles for testing
- pytest test structure to validate behavior

14. Deployment Considerations

- Docker + Docker Compose support for consistent environments
 - environment variables for OpenAI + LangSmith keys
 - mobile-friendly output rules (no emojis, structured JSON)
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Challenges and Learnings

15. Challenges

- Aligning many input formats (raw analytics keys vary) into consistent score extraction
- Balancing “fast feedback” (no LLM) with “high quality feedback” (LLM)
- Avoiding brittle prompts and ensuring the output is always structured JSON
- Designing trend logic that is meaningful while history is still mocked (hardcoded)

16. Key Learnings

- A tiered design is critical: **templates guarantee reliability**, LLM adds richness
 - Consistent schemas are as important as the model itself (the app depends on stability)
 - Trend-based feedback increases perceived value (players feel progress over time)
 - Drill recommendation mapping is a simple but powerful way to make feedback actionable
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Recommendations and Next Steps

17. Recommended Future Improvements

1. Replace hardcoded history with a real **database-backed history service**
 2. Expand drill database beyond Toe Taps (more coverage, more variety)
 3. Add authentication/rate limiting if deployed publicly
 4. Improve score extraction robustness for edge-case analytics payloads
 5. Add monitoring dashboards (request volume, response latency, error rate)
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18. Conclusion

This internship delivered value in two essential layers of the UPRO feedback pipeline:

- **Data Labeling:** created a structured dataset of JSON labeling exports for drill videos (primarily Toe Tap drills), uploaded to the shared Drive and ready for analytics usage.
- **AI Feedback System:** contributed to a production-ready feedback generator that standardizes metrics, supports multi-drill sessions, adds trend awareness, and provides age-appropriate coaching feedback through a robust FastAPI API.

The project evolved from early prototyping to a more scalable team approach. The final team solution is well-structured, testable, and aligned with deployment and product requirements—making it a strong foundation for real user integration.